22 **A**, **B** and **C** are three similar solids.

The surface area of **A** is 24 cm^2
The surface area of **B** is 54 cm^2

The height of **B** : the height of **C** = 3 : 7
The volume of **A** is 72 cm^3

C is made of wood with density 0.14 g/cm^3

Work out the mass of **C**.

Area ratio A : B = 24 : 54 = 4 : 9

Length ratio A : B = √4 : √9 = 2 : 3

1 mark
Square-root area ratio.

Length ratio B : C = 3 : 7

A	B	C
2	3	7

Length ratio A : C = 2 : 7

Volume ratio A : C = $2^3 : 7^3$ = 8 : 343

1 mark
Cube length ratio.

Volume of C = 72 × 343 ÷ 8 = 3087 cm^3

1 mark
Scale the volume.

Mass of C = density × volume

1 mark
Density × volume.

= 0.14 × 3087 = 432.18 g

1 mark
432.18 g.

…….. g

(Total for Question 22 is 5 marks)